1. Wrong Forcing Frequency

Used sin(t) instead of sin(ωt). This made the system assume low frequency while actual system was high frequency, forcing model to distort parameters.

2. Incorrect Electrical ODE

Equation was improperly scaled and structured, causing imbalance between mechanical and electrical physics constraints.

3. Incorrect Data Generation

Used steady-state sine approximation instead of solving ODE. Data did not match governing physics, leading to inconsistent learning.

4. Time Scaling Issue

High-frequency input made learning unstable. Neural network struggled with rapidly oscillating signals.

5. Missing Initial Conditions

ODE has infinite solutions without initial conditions. Model drifted due to lack of anchoring.

6. Loss Imbalance

Data loss dominated physics loss, causing model to prioritize fitting data instead of respecting physics.

7. Two Separate Networks

Using two models for same physics introduced instability and conflicting gradients.

8. Incorrect Use of Sin Activation

Used sine activation without proper scaling and initialization. Needed SIREN structure for stability.

9. Cp Scaling Issue

Mixed unit conversion inside training equation caused numerical instability and poor gradients.